11.3 The Real Pendulum

WORKSHEET

The question

Consider a simple pendulum of length l and angular displacement θ, which is **not** assumed to be small. Show that

$$\frac{1}{2}l\left(\frac{\mathrm{d}\theta}{\mathrm{d}t}\right)^2 = g(\cos\theta - \cos\gamma),$$

where γ is the maximum value of θ. Show also that the period P is given by

$$P = 2\sqrt{\frac{l}{g}}\int_0^{\gamma}\left(\sin^2(\gamma/2) - \sin^2(\theta/2)\right)^{-\frac{1}{2}}\,\mathrm{d}\theta.$$

By using the substitution $\sin(\theta/2) = \sin(\gamma/2)\sin\phi$, and then finding an approximate expression for the integrand using the binomial expansion, show that for small values of γ the period is approximately

$$2\pi\sqrt{\frac{l}{g}}\left(1 + \frac{\gamma^2}{16}\right).$$

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself The bob moves on a circle of radius l. What are its speed and its height (relative to the lowest point) in terms of θ and $\dot{\theta}$?

STAGE 2 The period as an integral

Ask yourself How can you write $\cos\theta - \cos\gamma$ in terms of half-angles? And how much of the period is spent going from $\theta = 0$ to $\theta = \gamma$?

STAGE 3 Removing the singularity

Ask yourself The integrand becomes infinite at $\theta = \gamma$. Write $k = \sin\frac{\gamma}{2}$ and substitute $\sin\frac{\theta}{2} = k\sin\phi$. What range of ϕ corresponds to $0 \le \theta \le \gamma$? What happens to the integrand?

STAGE 4 The first correction

Ask yourself For small k, expand $(1 - k^2\sin^2\phi)^{-1/2}$ by the binomial theorem. What is $\int_0^{\pi/2} \sin^2\phi \, \mathrm{d}\phi$?

Try this next

1 **A tethered particle.** (From STEP III, 2015.) A particle of mass m on a smooth horizontal rail is attached to a peg at distance a from the rail by an elastic string of natural length a and modulus λ. Starting with the string just taut, it is given speed v along the rail. Show that $\dot{x}^2 = v^2 - k^2\left(\sqrt{x^2+a^2} - a\right)^2$, where $k^2 = \frac{\lambda}{ma}$, find the greatest displacement, and show that for $v \ll ka$ the period is approximately $\sqrt{\frac{32a}{kv}}\int_0^1 \frac{\mathrm{d}u}{\sqrt{1-u^4}}$. Why does the period now *depend* on v, even for small oscillations?

2 **The next term.** Carry the binomial expansion one term further and show that $P \approx 2\pi\sqrt{\frac{l}{g}}\left(1 + \frac{\gamma^2}{16} + \frac{11\gamma^4}{3072}\right)$.

3 **The AGM by hand.** Use three steps of the AGM to estimate the period of a pendulum with amplitude $90°$, and compare with the value 1.1803 in the figure.

4 **Clocks.** A pendulum clock is regulated to keep perfect time with an amplitude of $4°$. If its amplitude increases to $5°$, how many seconds a day does it lose?

Further reading

- **David Morin,** ***Introduction to Classical Mechanics*** (Cambridge University Press), Chapter 4, which treats the large-amplitude pendulum.
- **Jonathan Borwein & Peter Borwein,** ***Pi and the AGM*** (Wiley). Gauss's discovery and its uses, including the fastest ways to compute π.
- **Dava Sobel,** ***Longitude*** (Fourth Estate). The story of timekeeping at sea, beyond the pendulum.

The full guided solution, and what the question leads to at university, is in Section 11.3:
https://beyondstep.keithmcnulty.org/ch11/real-pendulum.html